

openGUTS Report

Project:

GUTS_report_Libellula_depressa

Project file:

No project file saved or loaded

Project description (optional):

No project description available

Software version:

openGUTS - 1.2

Date of report creation:

23/07/2026 13:20:23

Calibration

Calibration input data

Data set 1

File: Data edited manually

Description (optional):

Control group: 'Control'

Survival data of input data set 1:

Time [d]	Control	T1	T2	T3	T4	T5	T6
0	79	20	20	20	20	20	19
0.042	79	20	20	20	20	20	19
0.083	78	19	20	20	19	20	15
0.16	75	19	19	17	18	19	12
0.33	75	19	18	17	18	18	0
1	69	17	17	15	16	13	0
1.208	67	16	17	15	16	13	0
2	59	15	15	12	16	6	0
3	54	13	11	11	12	5	0

Concentration data of input data set 1:

Time [d]	Control	T1	T2	T3	T4	T5	T6
0	0	3	10	30	100	300	1000
0.2	0	2.9	9.6	28.8	96.1	288.3	960.9
0.4	0	2.8	9.2	27.7	92.3	277	923.2
0.6	0	2.7	8.9	26.6	88.7	266.1	887.1
0.8	0	2.6	8.5	25.6	85.2	255.7	852.4
1	0	2.5	8.2	24.6	81.9	245.7	819
1.2	0	2.4	7.9	23.6	78.7	236.1	787
1.4	0	2.3	7.6	22.7	75.6	226.8	756.2
1.6	0	2.2	7.3	21.8	72.7	218	726.6
1.8	0	2.1	7	20.9	69.8	209.4	698.1

2	0	2	6.7	20.1	67.1	201.2	670.8
2.2	0	1.9	6.4	19.3	64.5	193.4	644.5
2.4	0	1.9	6.2	18.6	61.9	185.8	619.3
2.6	0	1.8	6	17.9	59.5	178.5	595.1
2.8	0	1.7	5.7	17.2	57.2	171.5	571.8
3	0	1.6	5.5	16.5	54.9	164.8	549.4

Calibration settings

Calibration parameter settings for GUTS-RED-SD:

Parameter	Fit	Min	Max	Scale
kd	Yes	0.001641	1096	Log
mw	Yes	0.0006219	990	Norm
hb	Yes	1E-6	0.25	Norm
bw	Yes	3.512E-5	334.2	Log
Fs	No	1	1	Norm

Calibration parameter settings for GUTS-RED-IT:

Parameter	Fit	Min	Max	Scale
kd	Yes	0.001641	57.8	Log
mw	Yes	0.0006219	1806	Log
hb	Yes	1E-6	0.25	Norm
bw	No	Inf	Inf	Norm
Fs	Yes	1.05	20	Log

Note:

Expert settings were used (calibration settings may have been changed).

Parameter ranges were changed manually.

Calibration results

Fitted parameters for GUTS-RED-SD:

Best fit parameter values and their 95% CI

kd: 10.53 (4.541 - 29.82)

mw: 195.1 (149.8 - 266.8)

hb: 0.1527 (0.1168 - 0.2026)

bw: 0.01097 (0.005693 - 0.04137)

* edge of 95% parameter CI has run into a boundary

(this may also affect CIs of other parameters)

Goodness of fit for calibration data (GUTS-RED-SD):

Model efficiency (NSE, r-square): 0.9419

Normalised root-means-square error (NRMSE): 6.784 %

Minus log-likelihood (MLL): 271.15

AIC: 550.29

Survival probability prediction error (SPPE) for each treatment:

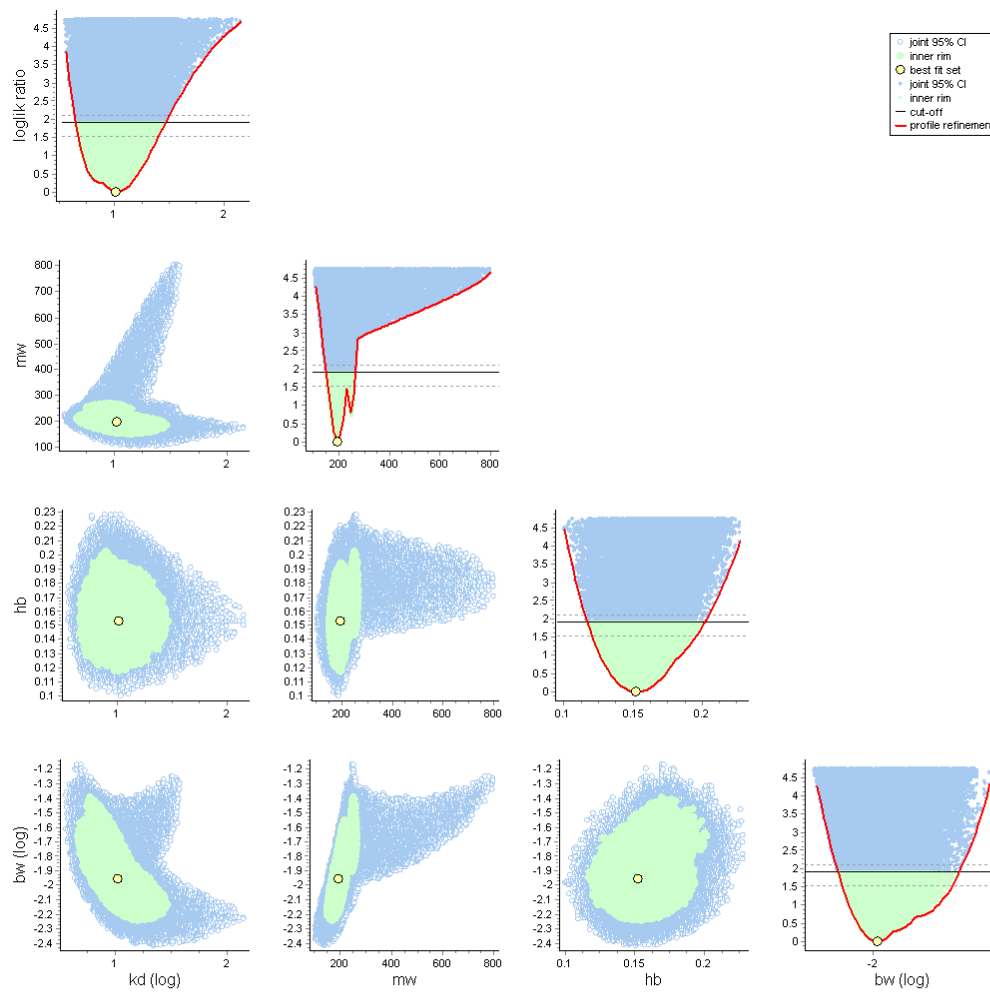
Data set	Treatment	Value
1	Control	5.114 %
1	T1	1.76 %
1	T2	-8.24 %
1	T3	-8.24 %
1	T4	-3.24 %
1	T5	2.551 %
1	T6	-1.2E-6 %

GUTS-RED-SD results table for LC_{x,t} [[C]], with 95% CI:

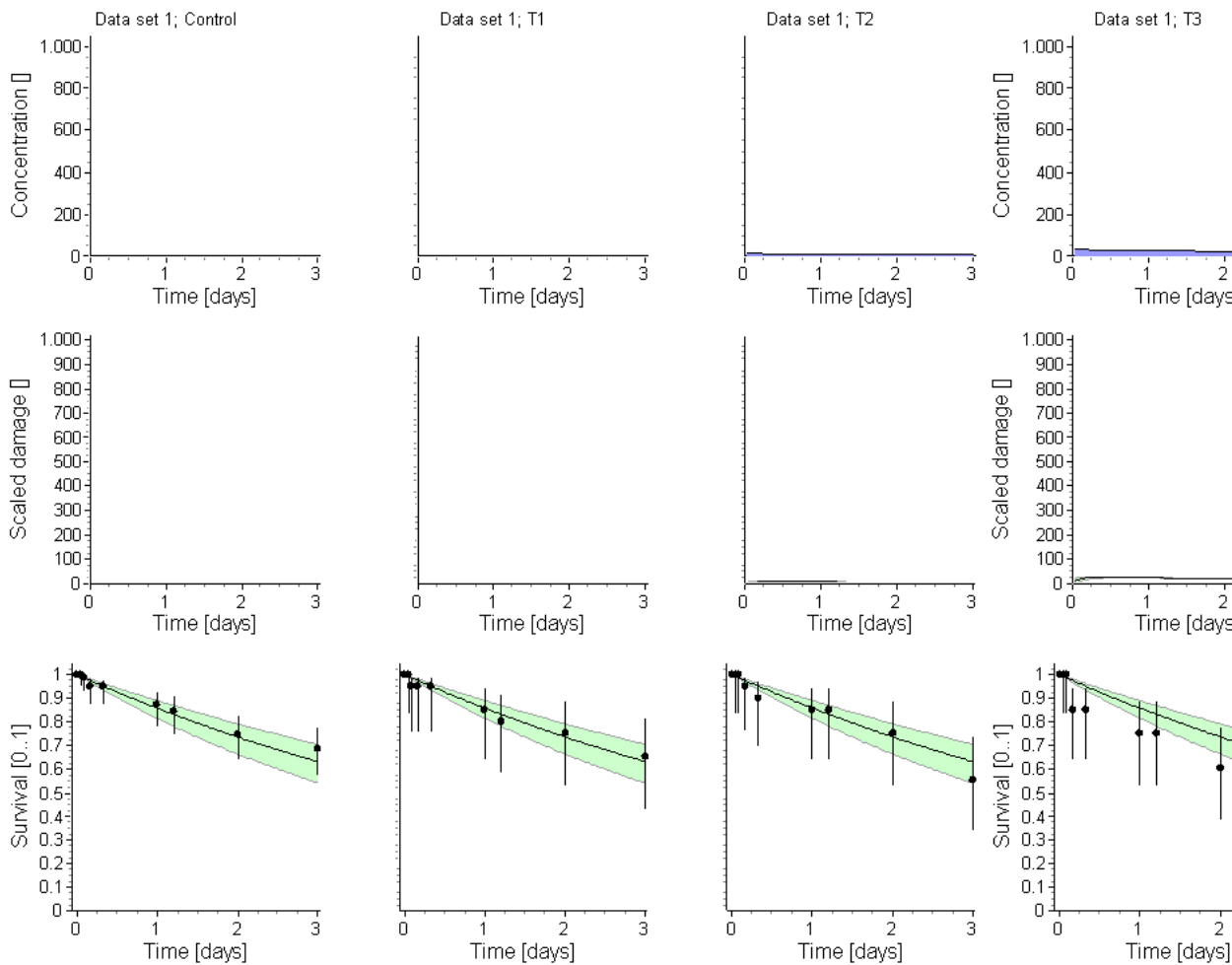
Time [d]	LC50	LC20	LC10
1	275.3 (244.4 - 325.8)	223.8 (187.1 - 287.1)	209.8 (168.1 - 276.7)
2	231.7 (198.5 - 291.8)	207.5 (166.3 - 274.4)	201.2 (156.7 - 270.8)
3	218.6 (182.5 - 282.1)	203 (159.6 - 271.7)	198.9 (153.3 - 269.5)

4	212.4 (173.9 - 277.6)	200.8 (156.4 - 270.5)	197.9 (151.8 - 268.8)
7	204.7 (162.6 - 272.5)	198.2 (152.4 - 269)	196.6 (149.9 - 268.1)
10	201.7 (158 - 270.9)	197.3 (150.9 - 268.5)	196.1 (149.1 - 267.9)
14	199.8 (154.9 - 269.8)	196.6 (149.9 - 268.1)	195.8 (148.6 - 267.7)
21	198.2 (152.4 - 269)	196.1 (149.1 - 267.9)	195.6 (148.3 - 267.6)
27	197.5 (151.3 - 268.6)	195.9 (148.7 - 267.7)	195.5 (148.1 - 267.5)
28	197.4 (151.2 - 268.6)	195.9 (148.7 - 267.7)	195.5 (148.1 - 267.5)
42	196.6 (149.9 - 268.1)	195.6 (148.3 - 267.6)	195.4 (147.9 - 267.5)
50	196.4 (149.6 - 268)	195.5 (148.2 - 267.6)	195.3 (147.8 - 267.4)
100	195.8 (148.5 - 267.7)	195.3 (147.8 - 267.4)	195.2 (147.7 - 267.4)
182	195.5 (148.1 - 267.5)	195.2 (147.7 - 267.4)	195.2 (147.6 - 267.4)

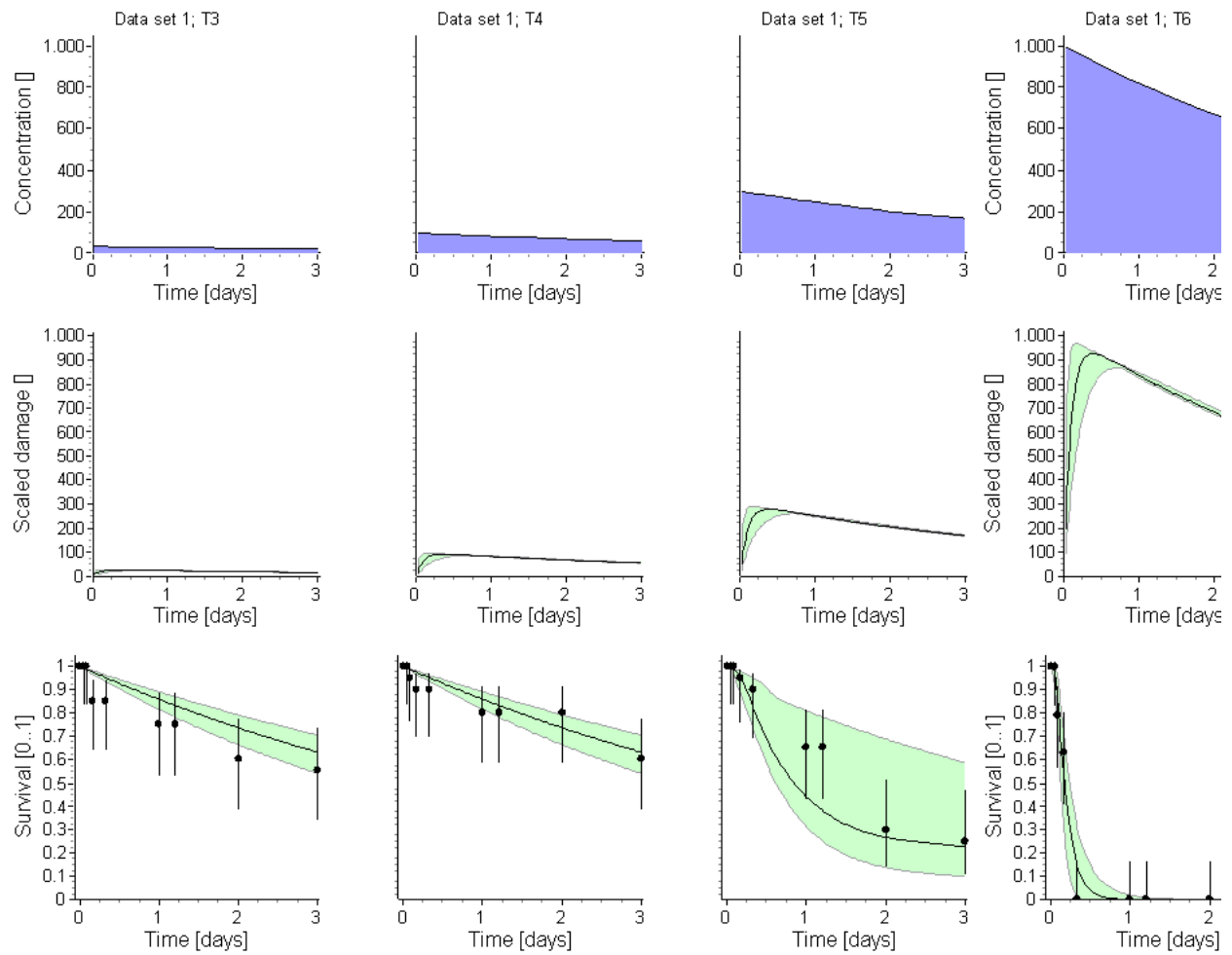
Parameter space plot for the calibration of GUTS-RED-SD:



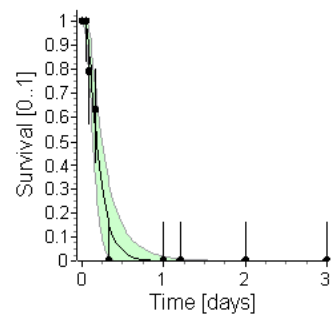
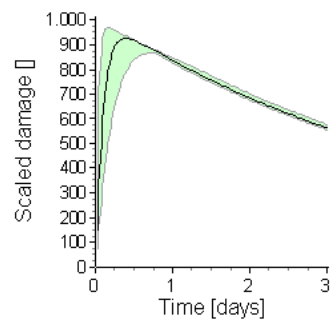
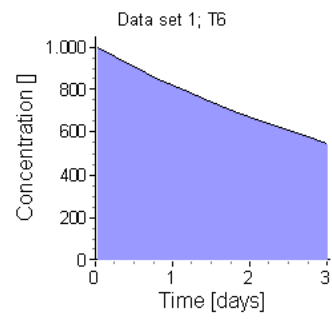
Exposure, damage and survival plots for the calibration of GUTS-RED-SD:



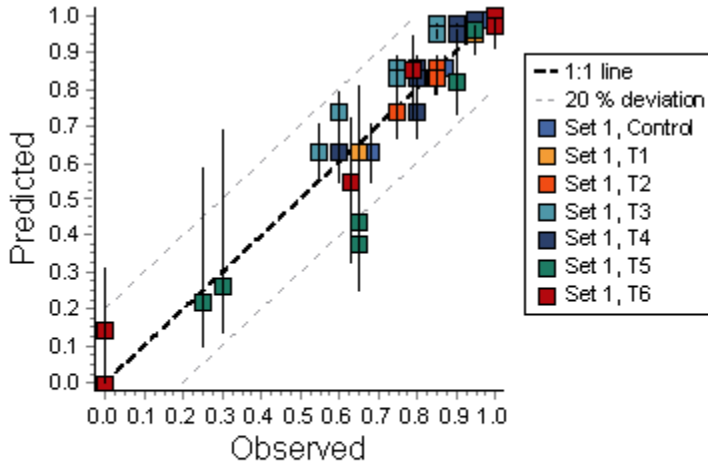
... continued plot:



... continued plot:



Observed vs. Predicted survival plot for the calibration of GUTS-RED-SD:



Observed vs. Predicted deaths plot for the calibration of GUTS-RED-SD:

LCx versus time with confidence intervals (plotted for 16 days, GUTS-RED-SD):

Fitted parameters for GUTS-RED-IT:

Best fit parameter values and their 95% CI

kd: 0.9784 (0.49 - 3.657)

mw: 174.8 (102.3 - 434.1)

hb: 0.1547 (0.1182 - 0.2029)

Fs: 2.651 (1.714 - 4.645)

* edge of 95% parameter CI has run into a boundary

(this may also affect CIs of other parameters)

Goodness of fit for calibration data (GUTS-RED-IT):

Model efficiency (NSE, r-square): 0.9437

Normalised root-means-square error (NRMSE): 6.769 %

Minus log-likelihood (MLL): 273.07

AIC: 554.14

Survival probability prediction error (SPPE) for each treatment:

Data set	Treatment	Value
1	Control	5.484 %
1	T1	2.13 %
1	T2	-7.87 %
1	T3	-7.85 %
1	T4	-1.25 %
1	T5	1.238 %
1	T6	-0.41 %

GUTS-RED-IT results table for LC_{x,t} [[C]], with 95% CI:

Time [d]	LC50	LC20	LC10
1	280.1 (230.6 - 449.5)	193.7 (141.5 - 324.9)	156.1 (102.2 - 276.3)
2	203.6 (153.6 - 435.2)	140.8 (93.17 - 313.8)	113.5 (68.05 - 265.1)
3	184.6 (128.3 - 434.8)	127.7 (77.76 - 313.4)	102.9 (56.68 - 264.7)
4	178.4 (115.9 - 434.8)	123.4 (70.52 - 313.4)	99.42 (51.46 - 264.7)
7	175 (103.4 - 434.8)	121 (63.34 - 313.4)	97.54 (46.55 - 264.7)

10	174.8 (100.9 - 434.8)	120.9 (61.84 - 313.4)	97.44 (45.61 - 264.7)
14	174.8 (100.3 - 434.8)	120.9 (61.48 - 313.4)	97.43 (45.39 - 264.7)
21	174.8 (100.2 - 434.8)	120.9 (61.42 - 313.4)	97.43 (45.37 - 264.7)
27	174.8 (100.2 - 434.8)	120.9 (61.42 - 313.4)	97.43 (45.36 - 264.7)
28	174.8 (100.2 - 434.8)	120.9 (61.42 - 313.4)	97.43 (45.36 - 264.7)
42	174.8 (100.2 - 434.8)	120.9 (61.42 - 313.4)	97.43 (45.36 - 264.7)
50	174.8 (100.2 - 434.8)	120.9 (61.42 - 313.4)	97.43 (45.36 - 264.7)
100	174.8 (100.2 - 434.8)	120.9 (61.42 - 313.4)	97.43 (45.36 - 264.7)
182	174.8 (100.2 - 434.8)	120.9 (61.42 - 313.4)	97.43 (45.36 - 264.7)

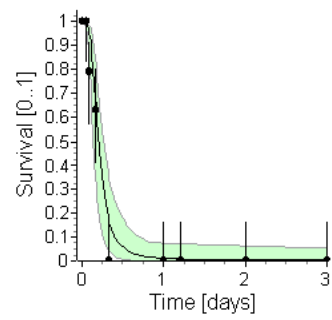
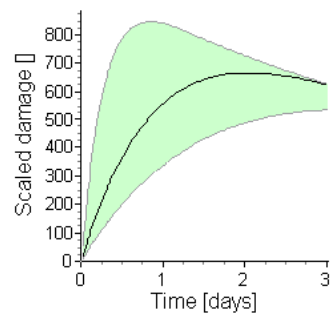
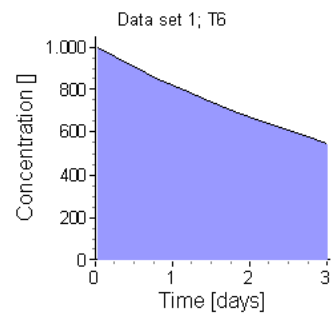
Plots for GUTS-RED-IT calibration:

Parameter space plot for the calibration of GUTS-RED-IT:

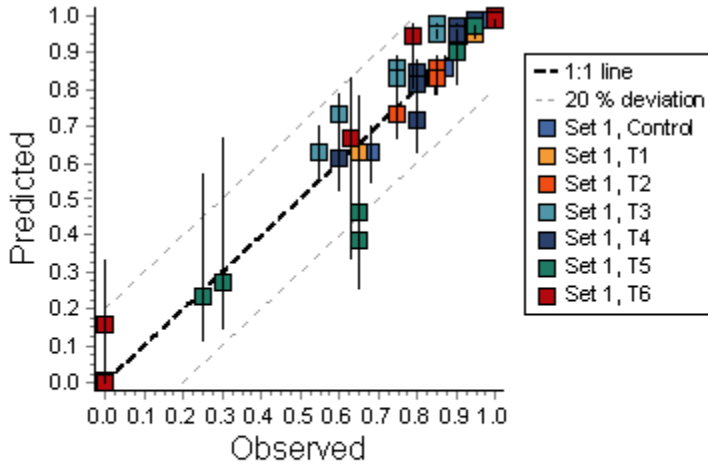
Exposure, damage and survival plots for the calibration of GUTS-RED-IT:

... continued plot:

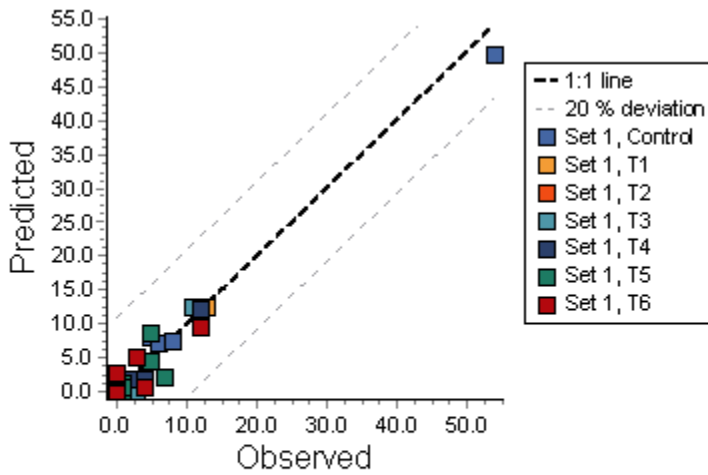
... continued plot:



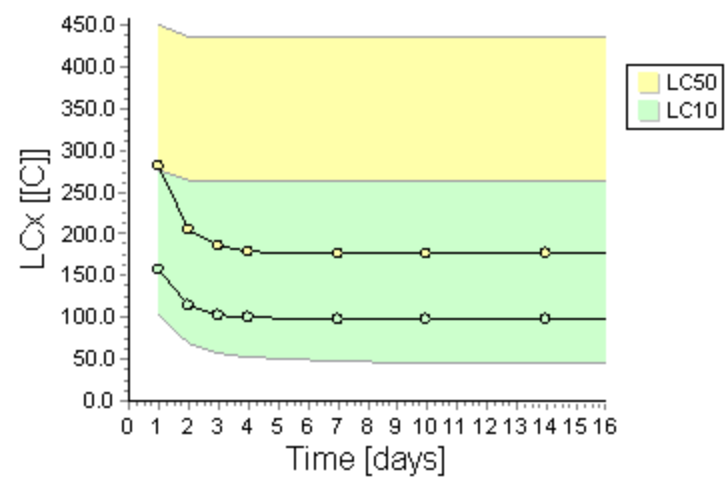
Observed vs. Predicted survival plot for the calibration of GUTS-RED-IT:



Observed vs. Predicted deaths plot for the calibration of GUTS-RED-IT:



LCx versus time with confidence intervals (plotted for 16 days, GUTS-RED-IT):



Validation

No validation performed!

Predictions

No predictions performed!